



Sensors and Instrumentation

Smart Greenhouse Monitor and Controller

Academic year: 2025-2026

Presented to: Dr. M. Aoudi

Prepared by:

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1. Contribution Summary

I contributed to the Smart Greenhouse Automation project by drafting the problem statement, designing the hardware circuit connections, and building a transitional path between hardware and software. I listed sensor connections for hardware teammates and explained sampling rates for software teammates based on physical dynamics (soil moisture every 3 s, DHT11 every 2 s, HX711 every 2 s, YF-S201 every 1 s, LDR every 200 ms, LCD every 3 s). I recorded raw sensor results, applied the logic of existing functions manually to convert values, derived calibration equations, and calculated errors before and after calibration. I prepared tables for absolute and relative errors, analyzed graphs, and collaborated integrated findings into the final report.

2. Evidence of Work Performed

- Drafted problem statement and system scope.

2. Problem Statement

Human observation and manual control are the traditional management technique used in greenhouses and they can result in inconsistent watering, lightning, and ventilation.

The problem to be addressed is how to maintain optimal and healthy growth conditions automatically using real time sensor feedback with a reliable and low-cost embedded system that monitors the variables and takes actions when thresholds are exceeded.

- Looked upon sensors types and helped in making the selection

Optical sensor	Light Sensor LDR 5mm		Measures ambient light to control grow lamp
			

- “Uploaded circuit schematic and explained hardware connections”

Component	Component Pin	Arduino Pin	Signal Type	Description / Function
YF-S201 Flow Sensor	Out (Yellow)	Pin 2	Digital Input (Interrupt)	Measures water flow volume via pulse count
2-Ch Relay Module	IN2 (Pump Control)	Pin 4	Digital Output	Controls the top relay channel to switch the 12V water pump
2-Ch Relay Module	IN1 (Fan Control)	Pin 5	Digital Output	Controls the bottom relay channel to switch the ventilation fan
Soil Moisture Sensor	VCC	Pin 6	Digital Output	Switched VCC power line to prevent probe corrosion

- **Defined sensor sampling rates based on physical dynamics**

- Soil moisture: every 3 s, soil water content changes slowly.
- Temperature & humidity (DHT11): every 2 s, balances sensor response time with environmental variation.

- **Recorded raw sensor data (soil moisture, LDR, HX711, DHT11, YF-S201).**

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--- SOIL SENSOR RAW CALIBRATION STARTING ---
RAW SENSOR VALUE: 1001
RAW SENSOR VALUE: 1001
RAW SENSOR VALUE: 1001

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1,786

2,789

3,788

4,784

5,783

6,787

7,790

8,787

9,782

10,784

11,789

12,788

13,784

Temperature Values

t1=33.0°C

t2=33.0°C

t3=33.0°C

t4=33.0°C

t5=33.0°C

t6=33.0°C

t7=33.0°C

t8=33.0°C

t9=33.0°C

t10=33.0°C

t11=33.0°C

t12=33.0°C

t13=33.0°C

t14=33.0°C

t15=33.0°C

t16=33.0°C

t17=33.0°C

t18=33.0°C

t19=33.0°C

t20=33.0°C

t21=33.0°C

Humidity results

at t1 = 64%

at t2 = 64%

at t3 = 64%

at t4 = 64%

at t5 = 64%

at t6 = 64%

at t7 = 64%

at t8 = 64%

at t9 = 63%

Figure 1 row values for light sensor

- **Derived calibration equations and applied manual conversions**

Using the two extreme points (20, 0 %) and (900, 50 %):

$$m_1 = \frac{50-0}{900-20} = \frac{50}{880} \sim 0.0568$$

$$b_1 = 0 - (0.0568 \cdot 20) \approx -1.14$$

So:

$$\text{Intensity (\%)} \approx 0.0568 \cdot \text{Raw} - 1.14$$

For the segment formed by (900,50) and (982,100)

$$I = m_2 \cdot \text{Raw} + b_2$$

$$m_2 = \frac{100-50}{982-900} = 50/82 \approx 0.6098$$

$$b_2 = 50 - (0.6098 \cdot 900)$$

$$I = 0.6098 \cdot \text{Raw} - 498.8 \text{ for } 900 \leq \text{Raw} \leq 982$$

$$I (\%) = \begin{cases} 0.0568 \cdot \text{Raw} - 1.1420 & \leq \text{Raw} \leq 900 \\ 0.6098 \cdot \text{Raw} - 498.8 & \leq \text{Raw} \leq 982 \end{cases}$$

Figure 2 calibration equation using linear regression

$\text{flowRateInLmin} = ((\text{float})\text{pulses}) / \text{flowCalibrationFactor};$

Raw values:

The volume of water collected is $k = \text{total pulses} / \text{collected volume}$

Live pulse count	20
Live pulse count	23
Live pulse count	26
Live pulse count	29
Live pulse count	32
Live pulse count	35

Figure 3 manual calculation of the calibrated values

- **Calculated absolute and relative errors before and after calibration for each of the five sensors.**

At random points:

Before:

- **Darkness (at t=1s Raw =786, mid range):**

$$I_{\text{raw}} = \frac{786}{1023} * 100 = 76.8\%$$

$$X_{\text{Reference}} = 50\% \rightarrow \text{Error} = \frac{|76.8 - 50| * 100}{50} = 53.6\%$$
- **Room light (at t=20s Raw =309, low range):** $I_{\text{raw}} = \frac{309}{1023} * 100 = 30.2\%$

$$X_{\text{Reference}} = 0\% \rightarrow \text{Error} = \text{absolute deviation} = 30.2\%$$
- **Flashlight (at t=42s, Raw=967):** $I_{\text{raw}} = \frac{967}{1023} * 100 = 94.5\%$

$$\text{Reference} = 100\% \rightarrow \text{Error} = 5.5\%$$

After Calibration

- **At 1 s (Raw = 786):** $I_{\text{cal}} = 0.0568 * 786 - 1.14 \approx 43.6\%$

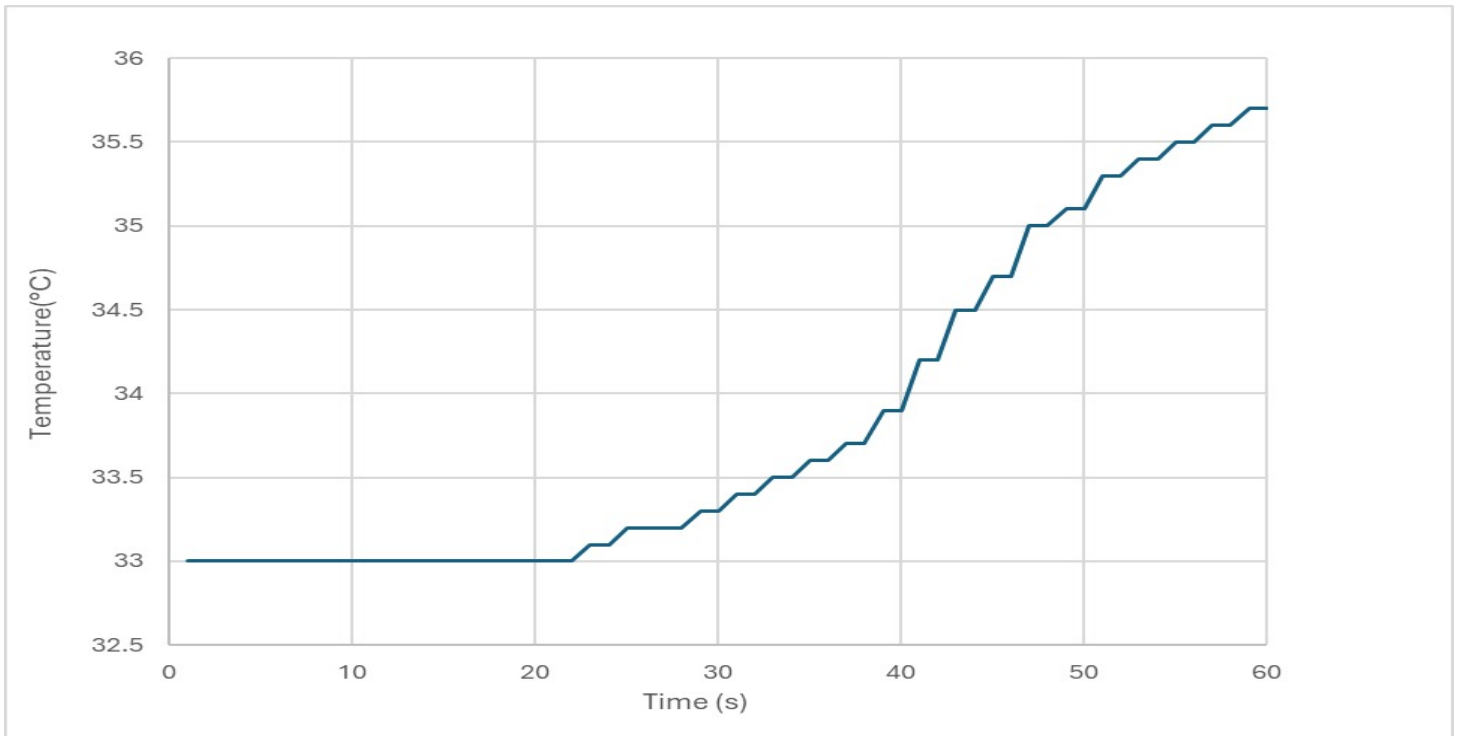
$$\text{Ref} = 50\% \rightarrow \text{Error} = 12.8\%$$
- **At 20 s (Raw = 309):** $I_{\text{cal}} = 0.0568 * 309 - 1.14 \approx 16.4\%$

$$\text{Ref} = 0\% \rightarrow \text{Error} = 16.4\%$$
- At 42 s (Raw = 967): $I_{\text{cal}} = 0.6098 * 967 - 498.8 \approx 90.6\%$
- $\text{Ref} = 100\% \rightarrow \text{Error} = 9.4\%$

Plotted calibration graphs



Time (s)	1	2	3	4	5	6	7
Temperature(°C)	33	33	33	33.3	33.9	35.1	35.7



- Collaborated with teammates to finalize calibration section

We cross-checked error values at reference points to confirm accuracy before finalizing the calibration section.

GitHub Commit History

- **Commit 1** Added *ProblemStatement.md* describing greenhouse automation challenge
- **Commit 2** Documented system architecture and hardware connections in *SystemArchitecture.md*
- **Commit 3** Defined sensor sampling rates based on physical dynamics in *SamplingRates.md*
- **Commit 4** Uploaded raw sensor data table for soil moisture, LDR, HX711, and DHT11
- **Commit 5** Derived calibration equations for soil moisture, LDR, DHT11, flow, and weight sensors
- **Commit 6** Calculated absolute and relative errors before and after calibration
- **Commit 7** Summarized calibration procedures, error analysis, and team collaboration

Personal reflection describing lessons learned.

Working on this project taught me valuable lessons both technically and personally. On the technical side, I gained deeper experience in sensor calibration, error analysis, and modular firmware design, which showed me how important accuracy and structure are in embedded systems. I learned how to organize data, apply calibration

equations, and document results clearly so that the system can be replicated and improved by others. On the personal side, I realized the importance of teamwork and communication—sharing tasks with my peers, trusting their contributions, and aligning our work through Git commits made the process smoother and more professional. I also discovered how meaningful it is to write clear commit messages and maintain a clean repository history, since these small details reflect discipline and responsibility. Overall, this project strengthened my technical confidence while also teaching me patience, collaboration, and the value of presenting work in a professional way.